

# Bank Management System

A database-driven banking desktop application built in Python with MySQL and Tkinter, supporting core banking operations through a graphical interface.

## Features

- Account creation, search, update, and deletion (full CRUD)
- Deposits and withdrawals with real-time balance updates
- Loan issuance with simple interest calculation, and loan repayment
- Login authentication to restrict access to the system
- Transaction data stored and retrieved via a relational MySQL schema

## Tech Stack

- Python (Tkinter for GUI)
- MySQL (relational schema for accounts and loans)
- `mysql-connector-python`

## Setup

1. Install MySQL and make sure it's running locally.
2. Install dependencies:

```
pip install mysql-connector-python
```

3. Set your MySQL password as an environment variable (do not hardcode it):

```
export DB_PASSWORD=your_password_here    # macOS/Linux
set DB_PASSWORD=your_password_here       # Windows cmd
```

4. Run the app:

```
python bank_system.py
```

## Design Notes

The database schema uses two linked tables — `Bank` and `Loan` — joined on account number, so loan status can be displayed alongside account details. Credentials are read from an environment variable rather than hardcoded, to avoid committing secrets to version control.